

Chapter 1 Introduction

In this chapter I will describe the main problems with existing computer scansion applications, and suggest a solution and ways to evaluate it against existing solutions.

Throughout the dissertation I will use the following symbols and abbreviations (the meanings are explained further in the text):

Symbol	Meaning
Stress	
w	Weakest stress
n	Low stress
m	Moderate stress/Secondary stress
s	Strongest stress/Primary stress
x	Metrically insignificant stress
Metrical Stress	
W	Weak stress
S	Strong stress
Musical Beat	
W	Weak beat
M	Moderate beat
S	Strong beat
Metrical Symbols	
	Foot break
	Caesura
Phonological Symbols	
\	Word break
<>	Intonational Unit (beginning and ending)
{ }	Phonological Phrase (beginning and ending)
[]	Clitic Phrase/Group (beginning and ending)
^	Polysyllabic Word (beginning and ending)
IU	Intonational Unit
PP	Phonological Phrase
CP	Clitic Phrase/Group
Scandroid Symbols (used by Hartman, 2005)	
/	Strong stress
	Foot break
x	Weak stress

Table 1: symbols used in the dissertation

1.1 Description of the problem domain

1.1.1 Scansion difficulties

Most poetry in English from C15 onwards has been composed to conform to a rhythmical pattern of stressed and unstressed syllables. For example, iambic pentameter poems are composed of lines which ideally have an unstressed syllable followed by a stressed one repeated five times – its metrical scheme (Table 2c). Scansion is the process of identifying the pattern of stresses in any given line (Table 2b). Classical Scansion matches the stressed/unstressed pattern of syllables to the ideal pattern for a given meter. It allows, to a limited degree, unstressed syllables where stressed are expected and, to a lesser degree, the reverse – in Table 2, for example, the stresses in “black hairs grow on” (sssw) do not match the expected metrical scheme (wsws). This method is by far the most popular form of scansion, and has recently been implemented in numerous computer analyses of poetry.

“If hairs be wires, black hairs grow on her head” (Shakespeare, Sonnet 130.4)		
a	Line	If hairs be wires, black hairs grow on her head
b	Stress pattern	If(w) hairs(s) be(w) wires(s), black(s) hairs(s) grow(s) on(w) her(w) head(s) ws ws ss sw ws
c	Metrical scheme	WSIWSIWSIWSIWS (Iambic Pentameter)

Table 2: Classical Scansion

Computer-based scansion is, in fact, increasingly desirable: given the decline in the teaching of scansion in English courses, and the complexity involved in matching stress patterns to metrical schemes, very few people, even literary experts, are equipped to appreciate what is arguably an essential element of most English poetry (Lieberman, 2004). Computer applications offer the possibility of providing accurate scansions, generated from objective principles, for whichever poem a user chooses, whether previously scanned or not. Expert scansions often disagree, and many are not explained

from first principles – a system which rigidly and transparently applies fixed principles would be a useful didactic tool, even if the end result was flawed. One which approximated reliable expert scansion would be even more useful. However, existing applications have significant deficiencies, especially when compared to improvements made to traditional scansion theory by linguists over the past forty years or more.

Existing computer scansion applications implement a range of scansion procedures. Traditional scansion is adopted by Raabe's program (1975), Bellin's LISP Scansion program (1997), Plamondon's *AnalysePoems* (2006) and one algorithm used in Hartman's *Scandroid* (2005). They all suffer from the shortcomings of the traditional method, for example, inadequate assessment of stress patterns and consequent inaccuracies in identifying meter. A more detailed discussion of these is provided below (1.1.2).

Dilligan and Lynn's program (1973) uses an early linguistic theory (Halle-Keyser) to identify whether lines are metrical or not. However, it does not produce the scansion of individual lines.

Hartman (2005) also uses a simplified version of this theory to derive scansion. However, it is limited to pentameter verse, biased towards iambic pentameter, and not very sophisticated in its application of the theory. Whilst the bias towards pentameter can be removed, Hartman himself acknowledges that his system is deficient in its assignment of stress, and subjective in its application of scansion.

Fabb's Bracketted Grid Theory (2003a, 2003b and 2006) imposes regular patterns on the line to determine which pattern is the closest fit, using the Generative Metrics concept of the stress maximum to anchor a line's stresses against the expected metrical stresses. It is unable to determine some meters correctly (for example, Catalectic Trochaic Trimeter).

Hayward's system (1996a and 1996b) references a number of linguistic concepts, but does not rigorously implement the theory behind them. It also requires user assessment of features of the line to work. Although the system uses a computer to process the interaction of stress-determining features of adjacent syllables, it does not use it to analyse the features themselves. The system is not a good candidate for full automation.

There is no program, nor any scansion procedure, which implements the phonological metrics theory of a core group of linguistic experts (Hayes, 2005). The theory has a relatively long history – it is derived from early observations by Magnuson and Ryder (1971) and Kiparsky (1977) that metrical stress matches word stress more strictly at the end of phrases than at the start, and is combined with an analysis of phrases based on phonology first formulated by Selkirk (1978). Its assumptions have been verified experimentally on numerous occasions: for example, Hayes (1984a), Hayes and Kaun (1996) and Tarlinskaja (1997). However, it has not been used by literature experts because it is perceived to be unwieldy compared to traditional scansion methods (Wallace, 1996 and Liberman, 2004).

1.1.2 Stress Assignment difficulties

In addition to scansion weaknesses, existing programs also suffer from inaccuracies in stress assignments. Since the stress pattern of a line forms the basis of scansion, errors at this level are extremely serious. Most programs assume only two levels of stress (for example, Scandroid). AnalysePoems assumes three. However, research over the past forty years has determined that even words which appear to have equal stress are differentiated under certain circumstances, some of which are relevant in poetry. In practice this means that at least four levels of stress may be necessary for determining rhythm accurately (Jespersen cited in Wallace, 1996).

Moreover, stress patterns in words interact with their syntactic function, producing different outputs in specific contexts. Some examples are given in the following discussion.

Since at least C16, it has been observed that the rules which determine when monosyllables receive stress are not very straightforward (Puttenham, 1569 and Brown, 1963). More recently, linguists have classified monosyllables into closed (stressed) or open (unstressed) categories (Hammond, 1991, Attridge, 1995 and Cooper, 1997). The closed class consists of content words – nouns, adjectives, verbs, adverbs and demonstrative pronouns. The open class consists of non-content words: determiners, prepositions, conjunctions, personal pronouns etc. (STELLA Project, n.d.). For example, ‘of’ is generally unstressed, whereas ‘can’ (noun or verb) is generally stressed.

Words with more than one syntactic function are stressed in some syntactic contexts, but unstressed in others. These fall into two categories. The first consists of synonyms which have different syntactic classes each with a distinct stress pattern. For example, Table 3a and b.

No	Word	First Stress Pattern	Second Stress Pattern
a	convict	the(w) con(s)vict(w) stays(s) [noun]	he(w) con(w)victs(s) me(w) [verb]
b	that	that(s) man(s) [demonstrative]	I(w) saw(s) that(w) it(w) was(s) [conjunction]
c	can	I(w) can(w) see(s) [modal verb]	I(w) can(s) [main verb]

Table 3: lexical stress dependent on syntax

The second consists of words which have different stress patterns depending on their precise syntactic functions. For example, in Table 3c, when ‘can’ functions as a modal verb it is generally unstressed, but when it is a main verb, it is stressed.

Linguistic theories like these have made stress assignment much more accurate. However, only one computer system references them – Hayward’s; but it requires user input to do it, and does not make use of linguistic theories systematically.

1.2 Phonological Scansion

None of the existing computer systems combines accurate stress assignment with a sophisticated scansion theory backed up by linguistic research. However, having synthesised a scansion procedure from research in phonological metrics, I hope to demonstrate its effectiveness in a computer application. Below, I assess it against three preliminary tests for the procedure which are difficult for scansion procedures generally.

1.2.1 Comparison with expert scansion

In Table 3 expert human scansion of five lines of poetry are compared to the scansion produced by this phonological scansion procedure and three recent systems by Plamondon (AnalysePoems), Hartman (Scandroid) and Fabb (Bracketted Grid Theory), to demonstrate which is closest to the expert scansion: a process I will replicate experimentally below (3.5.3). Scansions which match the expert scansion are coloured green.

System	Scansion	Comment
“If hairs be wires, black wires grow on her head” (Shakespeare, Sonnet 130.4)		
Raffel (1992)	If(w) hairs(s) be(w) wires(s), black(w) wires(s) grow(s) on(w) her(w) head(s). wslwslwslwslws	An expert scansion
Anstead et al. (1998)	wslwslwslwslws	A graduate’s scansion. It violates the natural rhythm of grow(s) on(w) by forcing a iambic rhythm to match the meter
AnalysePoems, Plamondon (2006)	if(0) hairs(2) be(0) WIRES(2) black(2)WIRES(2) grow(2) on(0) her(0) head(2) wslwslsslwslws	The scansion has an unmetrical series of 4 strong stresses. It does not take into account the break after “be wires”
Scandroid, Hartman (2005)	x / / / lx / lx / lx / wslsslwslwslws	The scansion has an unmetrical series of 3 strong stresses. It does not takes into account the break after “be wires”
Phonological Scansion	Xsw]s}n]sw]xws} = wslwsl-mslswlws	Matches the expert scansion. The m marks an intermediate stress which matches the lws pattern in the third foot
“Through Eden took their solitary way” (Milton, Paradise Lost 12.659)		
Milton (1674)	“English heroic verse without rhyme” i.e./ unrhymed iambic pentameter wslwslwslwslws	Expert scansion, in fact the author’s own assessment
Scandroid, Hartman (2005)	/ / x / x x /lx xl / sswswlswlwwls	The Scandroid gave up trying to scan this line – it found no significant series of iambs based on inadequate stress assignments.
Bracketted Grid Theory, Fabb (2003b)	wslwslwslwslws	Matches expert scansion
Phonological Scansion	Xsw}s}xswsw]s} = wslwslwslwslws	Matches Fabb’s analysis and expert scansion

System	Scansion	Comment
“There they are my fifty men and women” (Browning, One More Word 1)		
Halpern (1962)	“Browning’s trochaic pentameter poem One Word More” i.e./ swlswlswlswlsw	An expert opinion
Scandroid, Hartman (2005)	x / / / x / x / x / x wslswlswlswlsw	The Scandroid tries to force a iambic reading on a trochaic verse
Phonological Scansion	xnm>xsw]s}x[sw]> = snlmwslswlswlsw	produces a regular trochaic pentameter scansion
“What had I given to hear the soft sweep” (Charlotte Bronte, Diving 9)		
Bracketted Grid Theory, Fabb (2003b)	slwwslwwslwsw	I could not find an expert scansion for this verse, but it is probably dactylic trimeter as Fabb suggests
Scandroid, Hartman (2005)	x / x / x x / / x / / / wslswlwwslwslss	The Scandroid tries to scan the anapaests as iambs, even when forced to look for anapaests. Inaccurate stress assignment, and a failure to prioritise the end of the phrases produces the error
Phonological Scansion	s]xxsw]ws}xms} = slwwslwwslwms	Matches Fabb’s scansion (and probably the intended meter)
“Gone were but the winter” (Christina Rossetti, Spring Quiet 1)		
Fussell (1979)	swlswlsw	Expert scansion cited in Fabb (2006)
Bracketted Grid Theory, Fabb (2006)	wwslwsw	Fabb scans the entire poem with only two feet per line
Scandroid, Hartman (2005)	/ / x x / x / x slwwlswlsw	The Scandroid fails to find a significant series of iambs and gives up the scansion. Inadequate stress assignment is partly to blame
Phonological Scansion	Sw]/xxsw} = swlnwslsw	Matches the expert scansion
Key Expert scansions Scansion matching expert scansion		

Table 4: test verses for computer scansion systems

The table shows that the Phonological Scansion process is the only one to consistently match the expert scansion for the chosen lines.

1.2.2 Identification of rhythm in ambiguous stress patterns

Phonological scansion is also able to reliably identify the poet's intended rhythm in lines whose stress pattern fits two or more metrical patterns. For example, the stress pattern swswws can be scanned as iambic (-slwslwslw) or trochaic (swslwslwsl-). Atkins (1942) called this "the great metrical riddle on which so much time and so much ink has been expended". Tsur (1998) and Tarlinskaja (1993) conclude that the only reliable way to identify which is the correct rhythm is by a comparison with previous verses, although Hascall (1971) and Wright (1988) note that the rhythms of individual phrases or words help sensitive critics identify the meter within verses (supported by Atkins, 1942, Attridge, 1982, Fabb, 2002 and Wheatley, 2006 following Cureton, 1996). Relying on this assumption, the phonological scansion procedure is able to identify meter without reference to other verses by identifying the rhythm of polysyllabic words – see

Table 5 which analyses key lines from the poem "Epitaph on the Countess Dowager of Pembroke".

Scansion type	Analysis	Comment
Willam Browne, “Epitaph on the Countess Dowager of Pembroke”, lines 2 and 6		
Overall stress pattern	s w s w s w s	Either catalectic trochaic or headless iambic tetrameter
Line 2	Lies the subject of all verse	Hascall’s scansion is catalectic trochaic
Stress pattern	Lies(s) the(w) sub(s)ject(w) of(n) all(m) verse(s)	
Phonological scansion (meter)	xxsw}xms>	
Phonological scansion (rhythm)	(iambic=none, trochaic=‘sub(s)ject(w)’)	One polysyllabic word is trochaic. There are no iambic rhythms. So, the verse rhythm is trochaic.
Overall Scansion	swlswlnmls-	catalectic trochaic
Line 6	Time shall throw a dart at thee	Hascall’s scansion is headless iambic
Stress pattern	Time(s) shall(m) throw(s) a(w) dart(s) at(n) thee(m)	
Phonological scansion (meter)	xms}ws}nm>	
Phonological scansion (rhythm)	(iambic=‘shall(m) throw(s)’, ‘a(w) dart(s)’, trochaic=none)	One phrasal verb, and one end of phrase are iambic. There are no trochaic rhythms. The verse rhythm is iambic.
Overall Scansion	-slmslwslns	headless iambic
Key		
Scansion matching expert scansion		

Table 5: phonological scansion of rhythm (after Hascall, 1971)

Hascall (1971) notes it took “an exceptionally sensitive and knowledgeable critic” to realise that the poem’s first three lines are trochaic and the last three iambic. The phonological scansion procedure is able to replicate this sensitivity.

1.2.3 Assessment of Phonological Scansion procedure

In order to assess how effective the Phonological Scansion procedure is, I have classified verses according to difficulty. The procedure should be able to identify meter and scansion of the following categories of verses with decreasing accuracy, but still approximate expert scansion better than Scandroid and AnalysePoems:

No.	Category	Example
1	Verses of varying numbers of feet and types of feet (iambic, trochaic, dactylic and anapaestic) with no irregular feet	Iambic Pentameter: eg/ Shakespeare Sonnet 130.3 “If snow be white, why then her breasts are dun” Trochaic Trimeter: eg/C. Rossetti Spring Quiet 1 “Gone were but the winter”
2	Verses of varying number and type of feet with irregular scansion in phonologically insignificant syllables	Shakespeare Sonnet 130.2 “My mistress’ eyes are nothing like the sun” (Iambic Pentameter)
3	Verses written in Sapphics, Alcaics (fixed syllable meter)	Sapphics: eg/ Swinburne Sapphics (3x Sw/sx/sw/ws/ws/s, 1x Sw/ws/s) Alcaics: eg/ Tennyson, Milton (2x s/sw/ss/sww/sx/s, 1x s/sw/ss/sw/sx, 1x sww/sww/sw/sx)
4	Verses of varying number and type of feet with irregular scansion in one phonologically significant syllable (including headless lines and feminine endings)	Shakespeare Sonnet 130.1 “Coral is far more red than her lips’ red” (Iambic Pentameter)
5	Verses of varying number and type of feet with irregular scansion in more than one phonologically significant syllable.	C. Bronte Diving 1 “What had I given to hear the soft sweep” (Anapaestic Tetrameter)
6	Verses of varying number and type of feet where, for phonologically significant syllables, the number showing irregular scansion is equal to or greater than the number showing regular scansion	Browning Lost Leader 2.10 “There would be doubt hesitation and pain” (Dactylic tetrameter)
7	Free verse showing some regular feet	Frost Out, Out
8	Free verse without regular feet	Whitman Out of the cradle rocking endlessly

Table 6: categories to assess computer implementation of phonological scansion

T. Criteria for success of the theory

I have defined the following objectives to measure the theory’s success rate:

- T1. For categories 1-6 the procedure should match the meter identified by experts, and produce a scansion which is close or identical to expert scansion in the majority of cases (95%).
- T2. For categories 7-8, which are notoriously difficult to analyse, the procedure is less likely to succeed and should match 70% of scansions.
- T3. It should also either match or improve the competing computer system scansions.

In addition, Hayes (2005) has implemented an algorithm which ‘textsets’ (matching lexical stresses in lyrics to beats in the music, similar to scansion): it replicates human text-setting 71% of the time.

- T4. A procedure which replicates expert scansion better than 71% of the time would be a good result for the research.

1.3 Contribution to knowledge

The research tests whether a system employing linguistic theories of stress assignment and scansion makes an improvement in the identification of scansion by existing computer systems.

Over the past few years there has been an increase in computer scansion applications. This has been facilitated by the ease of implementing and publishing applications, but is probably motivated by a desire to make scansion more accessible, countering a perceived decline in the use and knowledge of scansion by students of English -compare Thomas (1913) with Walsh (1963), Woods et al. (1982) and Liberman (2004). However, the most popular applications use theories of scansion which do not fully